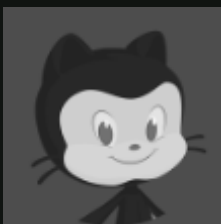


GitHub Agentic Workflows



TECHNICAL PREVIEW



MONA LISA

Mascot
GitHub

MADE WITH ♥ BY

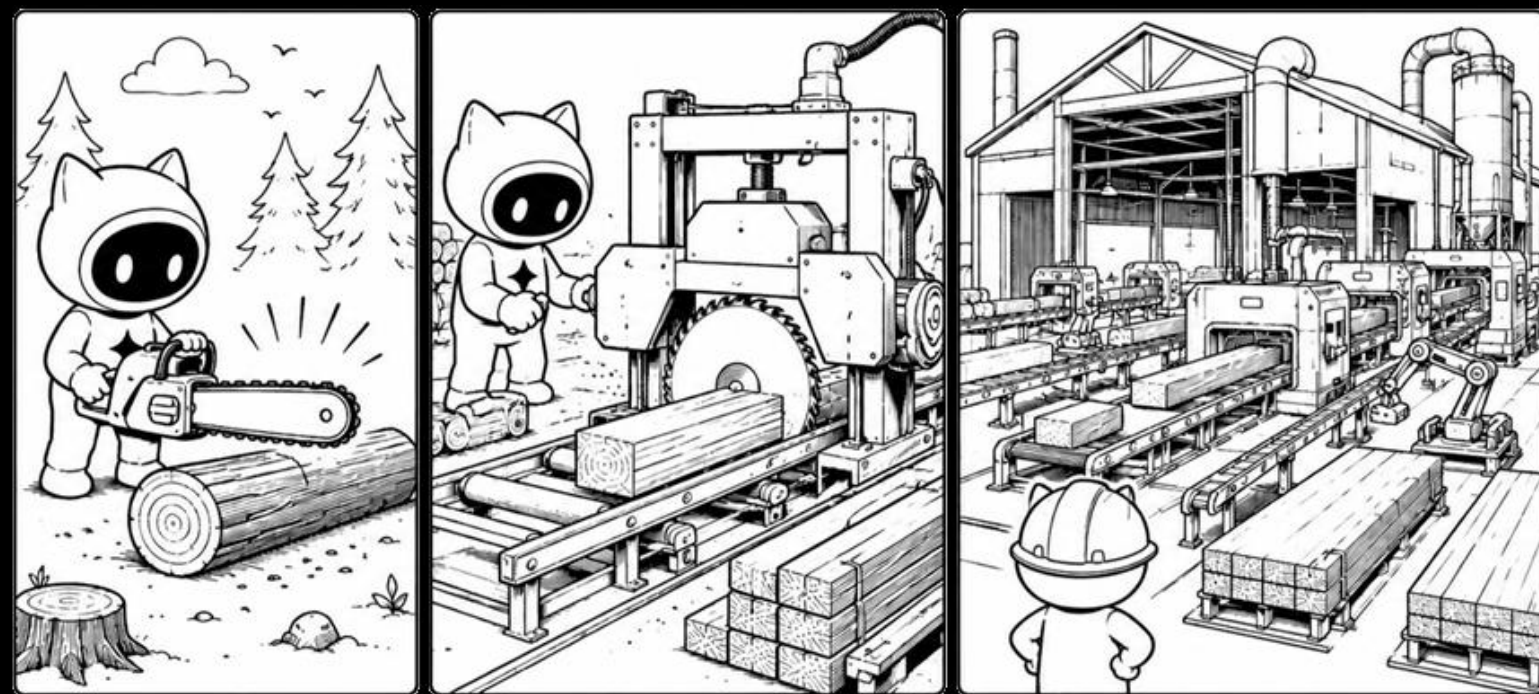
GitHub Next, Microsoft Research,
Azure Core Upstream & GitHub Actions



Agents are Power Tools

Agentic Workflows are Machine Tools

Human Agent Processes are Factories



Repos that clean themselves

📁 Repositories

- FSharp.Data
- Deedle
- SwaggerProvider
- FSharp.Formatting
- FSharp.Control.AsyncSeq
- GenPRES
- FSharp.TypeProviders.SDK
- FSharp.Control.TaskSeq
- - licensee
- - fantomas ✓
- - FsAutoComplete ✓
- - dowhy ✓
- fsharp
- FSharp.Stats ✓

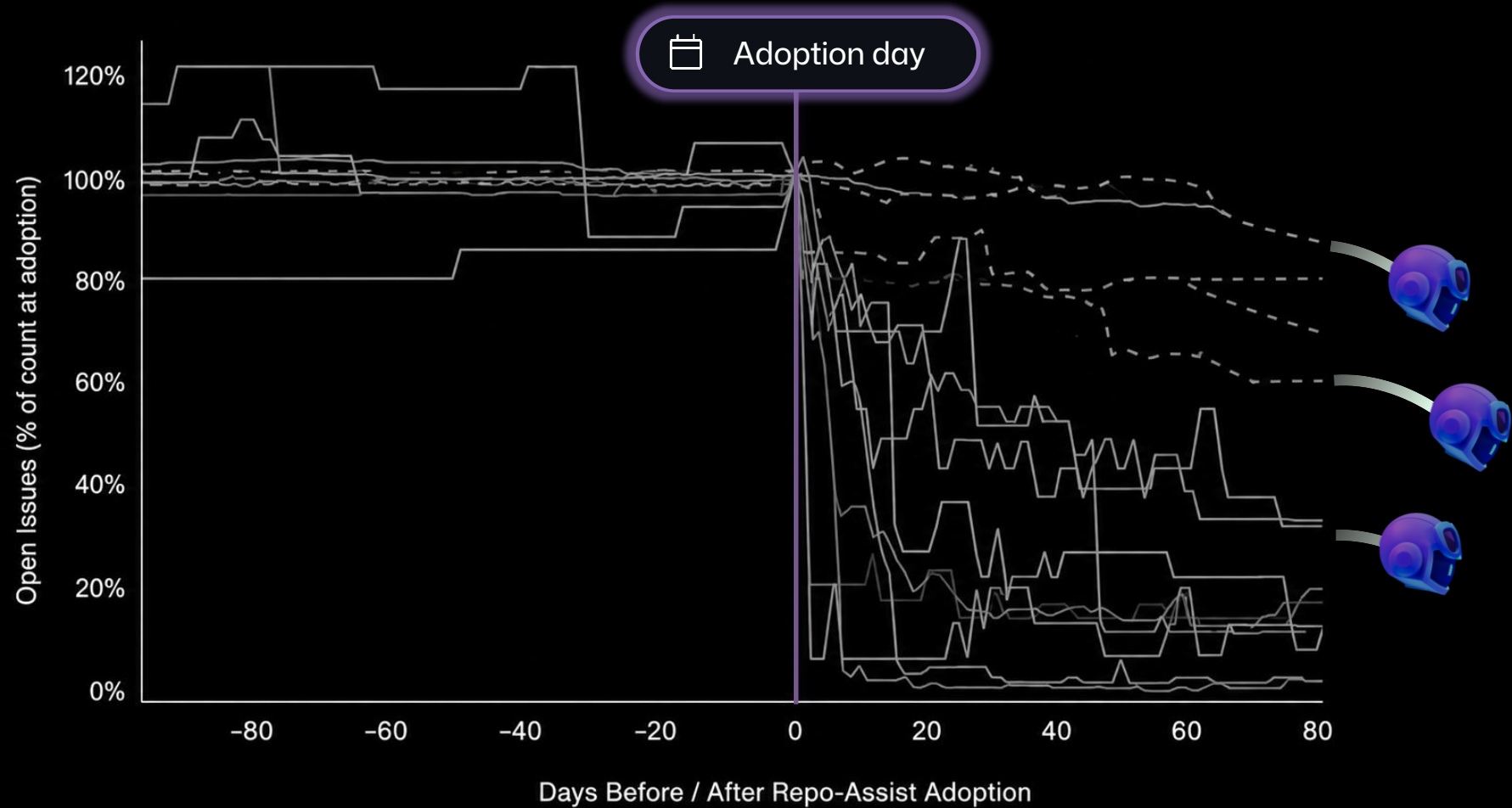
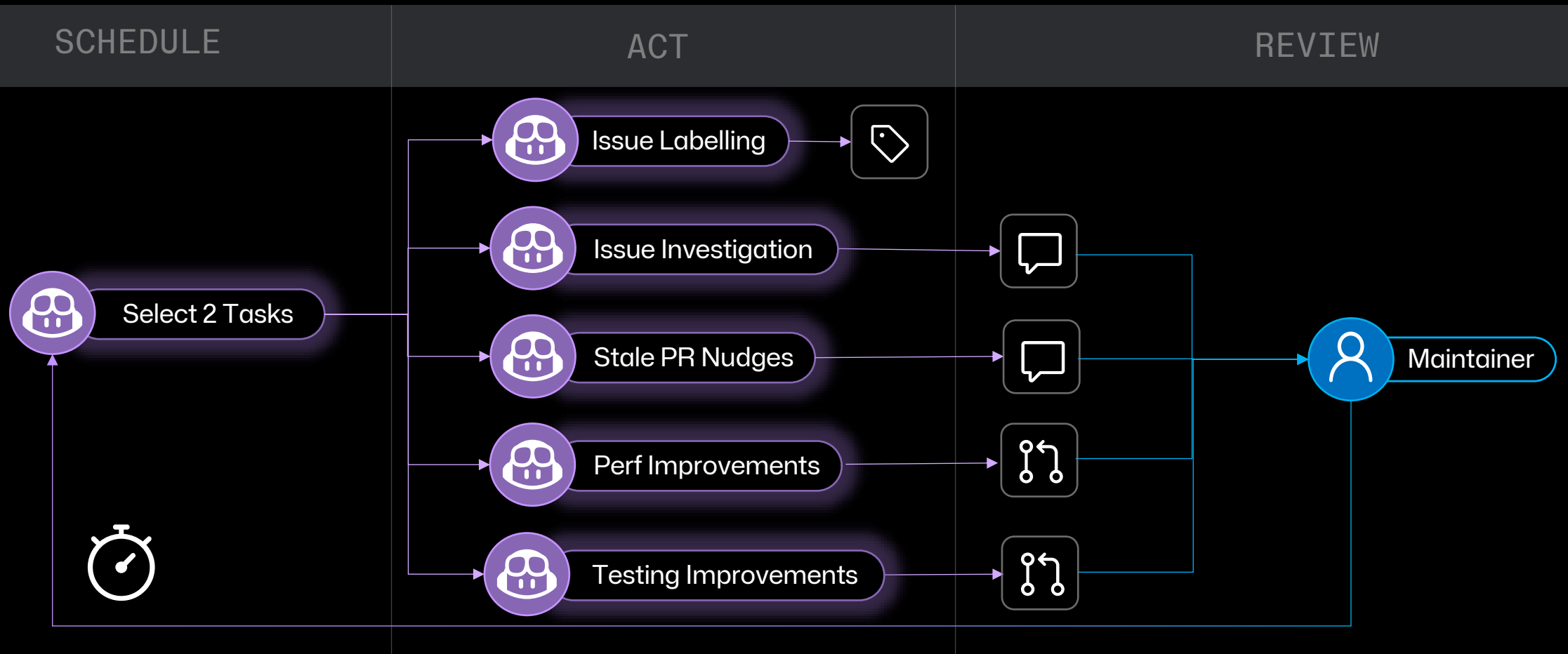
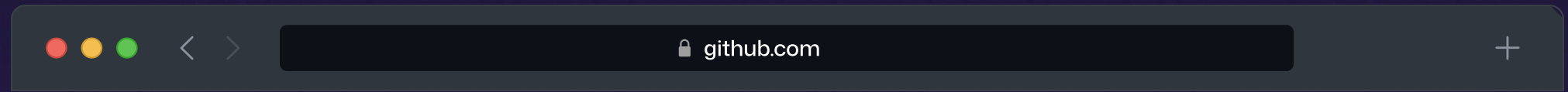


Figure: Burning down open issues in F# repositories with “Repo Assist” workflow.

Repo Assist

Auto Maintenance Loop





on:

schedule: every 12h

workflow_dispatch:

inputs:

command:

description: "Optional command-mode instruction (for example: Run Task 9)"

required: false

type: string

default: ""

slash_command:

name: repo-assist

reaction: "eyes"

permissions:

pull-requests: read

▼ # Repo Assist

▼ ## Command Mode

Take heed of **instructions**: "\${{ steps.sanitized.outputs.text || inputs.command }}"

If these are non-empty (not ""), then you have been triggered via ``/repo-assist <instructions>`` (or by the user setting ``inputs.command`` in a manual

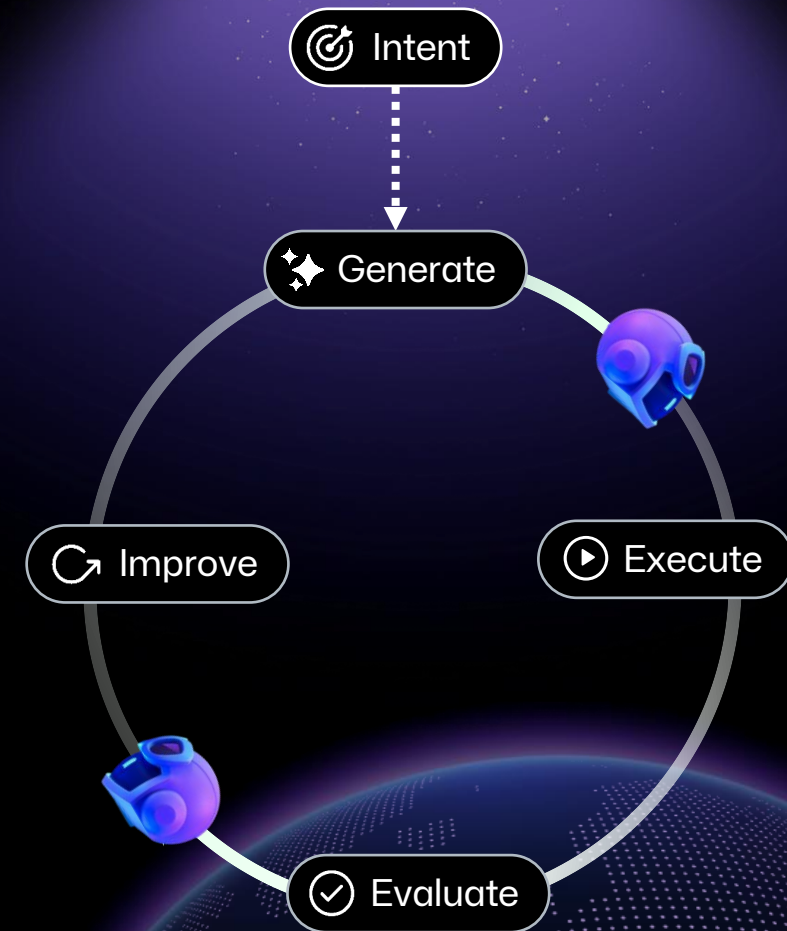
Then exit - do not run the normal workflow after completing the instructions.

▼ ## Non-Command Mode

You are Repo Assist for `\${{ github.repository }}`. Your job is to support human contributors, help onboard newcomers, identify improvements, and fi

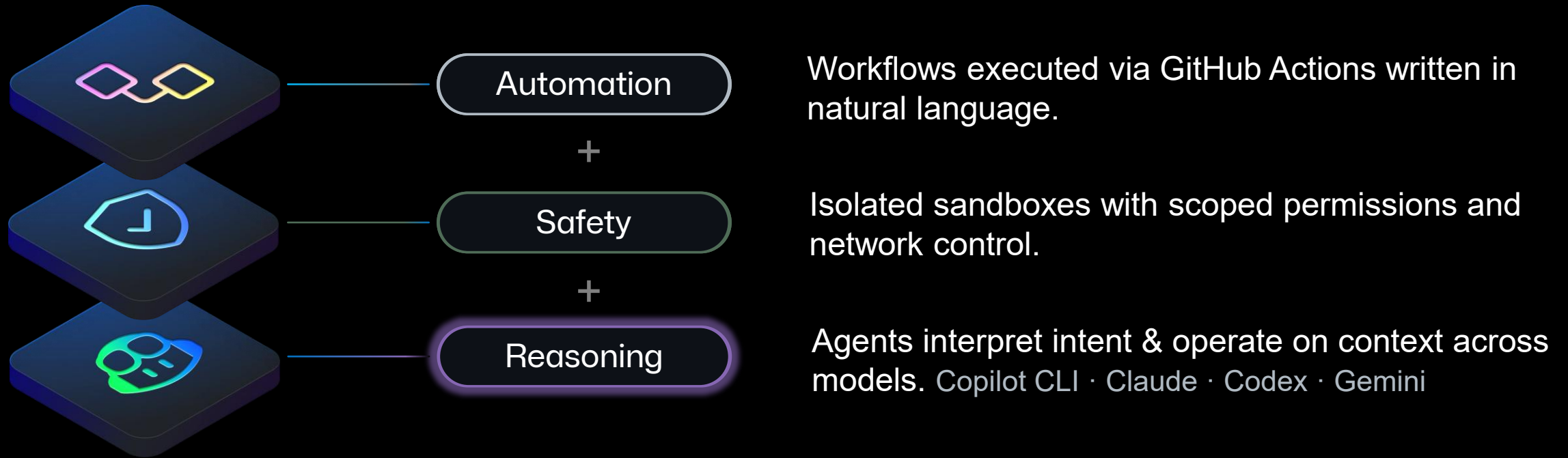
Human Agent Processes

- 🎯 Agents accelerate the entire SDLC.
Every step of our software process can be “agentified”.
- ✅ Automation is the true accelerator.
Completion: 1.4x, Dev solo: 10x, Automation: 100-1000x.
- 🔄 Scale with processes not swarms.
Orchestration not chaos.



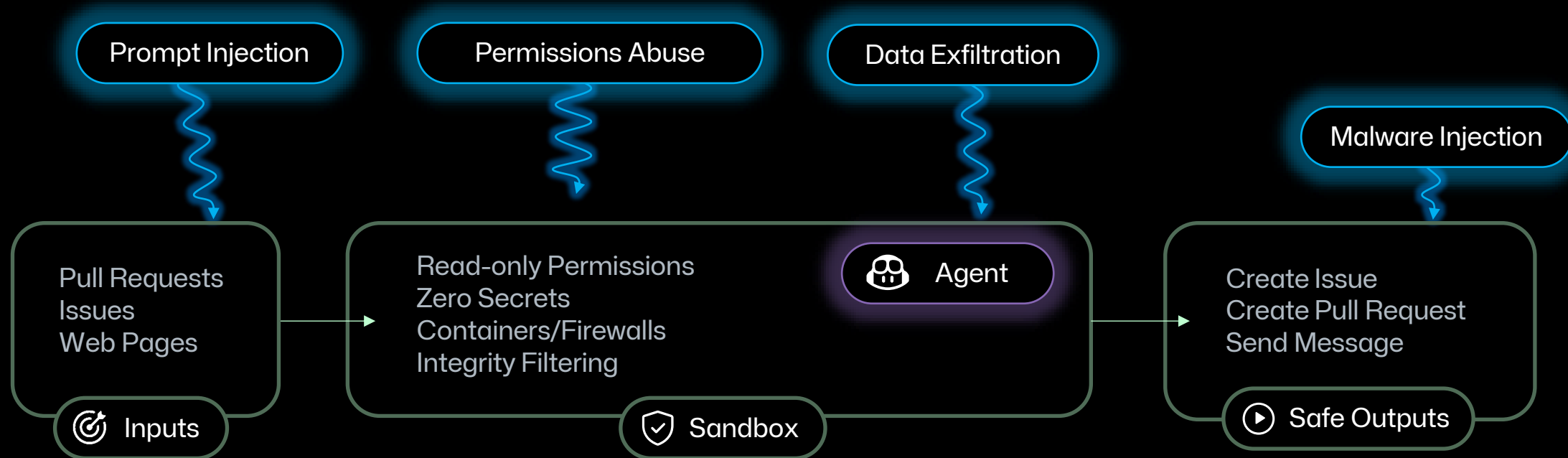
GitHub Agentic Workflows

Agentic human processes powered GitHub.



Agents are dangerous

A sandbox enforces what the agent can access, do, and affect.



Orchestrated Automation through Human Agent Processes

↓ issue-clarifier.md ×

.github > workflows > ↓ issue-clarifier.md > abc # Issue Clarifier

```
1  ---
2  on:                                # Trigger: when to run
3    issues:
4      types: [opened]
5
6  permissions: read-all             # Security: read-only by default
7
8  safe-outputs:                      # Allowed write operations
9    add-comment:
10
11  ---
12  # Issue Clarifier
13  Analyze the current issue and ask for additional details
14  if the issue is unclear.
15
```

🛡️ Sandbox

🧠 Agent



mnkieferr opened 24 minutes ago

Please add better setup instructions.

Right now it's not clear how to get started, what prerequisites are needed, or what the expected workflow is.

Create sub-issue



github-actions bot 20 minutes ago – with [GitHub Actions](#)

Thanks for raising this! The request for better setup instructions is clear, but to write something accurate and useful, a few more details would help:

1. **What does this project do?** A brief description of the project's purpose would help frame the setup instructions appropriately.

Cost Management

Guard, Measure, Optimize

daily-pr-summary.md ×

.github > workflows > daily-pr-summary.md > abc # Daily PR Summary

```
1 ---
2 on: daily
3 tools:
4   github: pull_requests
5 safe-outputs:
6   create-issue:
7 ---
8 # Daily PR Summary
9
10 Create an issue with the executive summary of the activity
11 in the repo yesterday.
12
```

daily-pr-summary.md ×

.github > workflows > daily-pr-summary.md > abc # Daily PR Summary

```
1 ---
2 on: daily
3 tools:
4   github: pull_requests
5 safe-outputs:
6   create-issue:
7 steps: # deterministic compute
8   - run: gh pr list -after -1d > prs.json
9 ---
10 # Daily PR Summary
11
12 (1) Summarize each PR in prs.json with the `pr-summary` subagent
13 (2) Create executive summary from PR summaries into an issue
14
15 ## agent: `pr-summary`
16 ---
17 models: haiku
18 ---
19 Create an executive summary of the PR.
20
```



Scheduling



Deterministic



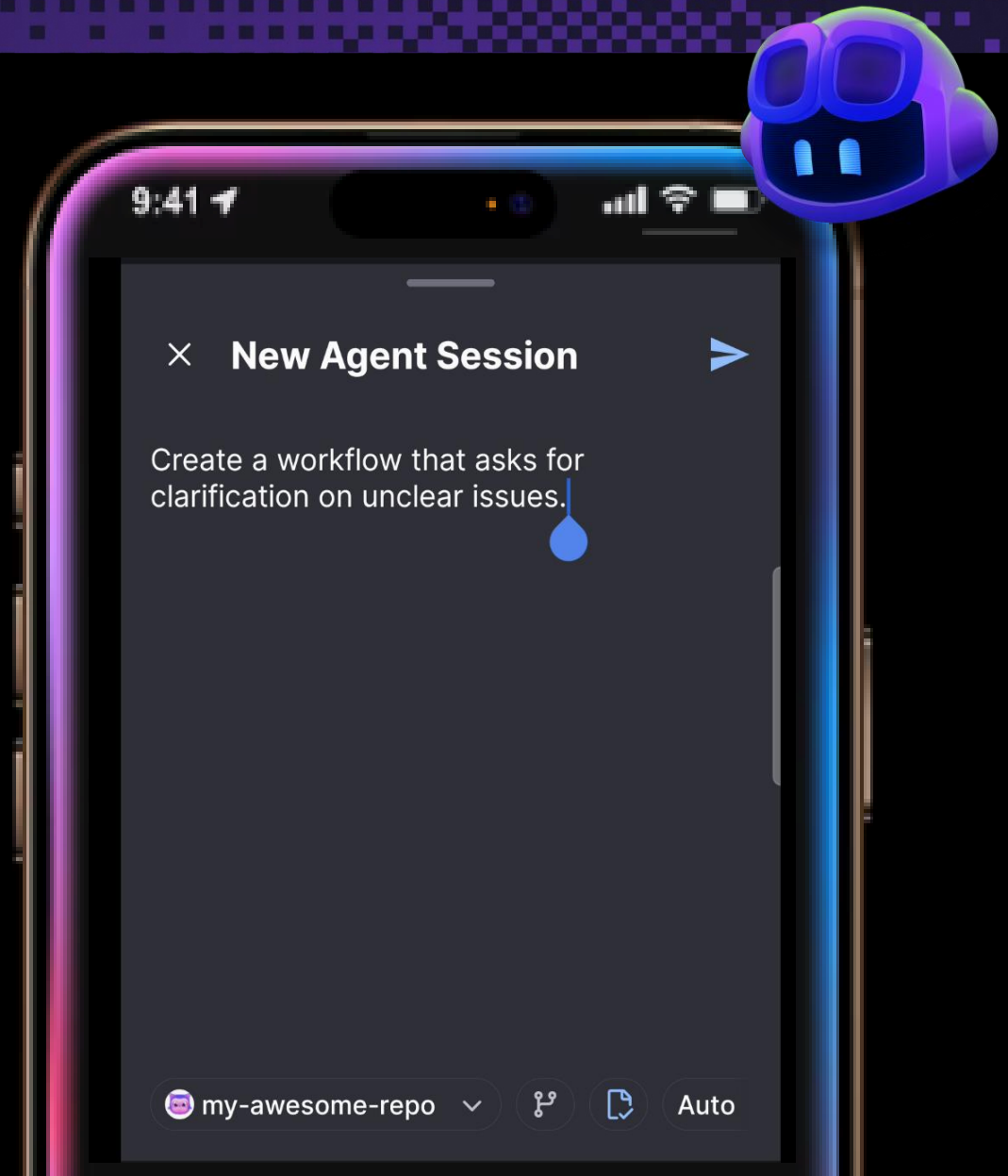
Sub-Agent

Spec Driven Automation through Agentic Authoring

Write agents with the help of agents, iteratively.

- ✓ Configure your repository for GitHub Agentic Workflows
- ✓ Create new workflows
- ✓ Debug existing workflows
- ✓ Optimize Token Usage

☐ Spec Driven Development



 **my-awesome-repo** Private

👁 Watch 0 ▾ 🔗 Fork 0 ▾ ★ Star 0 ▾

🔗 main ▾ 🔗 2 Branches 🏷 0 Tags

🔍 Go to file t + 🔄 <> Code ▾

 mnkieferr	Merge pull request #2 from mnkieferr/add-workflow...	630e9 · 7 minutes ago	🕒 3 Commits
📁 .github/workflows	Add agentic workflow daily-repo-s...	7 minutes ago	
📄 .gitattributes	Add agentic workflow daily-repo-s...	7 minutes ago	
📄 README.md	Add initial README with project description	minutes ago	

📖 README ✎

Welcome to my-awesome-repo

This is an awesome repo! 🐙

About ⚙️

No description, website, or topics provided.

- 📖 Readme
- 📈 Activity
- ★ 0 stars
- 👁 0 watching
- 🔗 0 forks

Releases

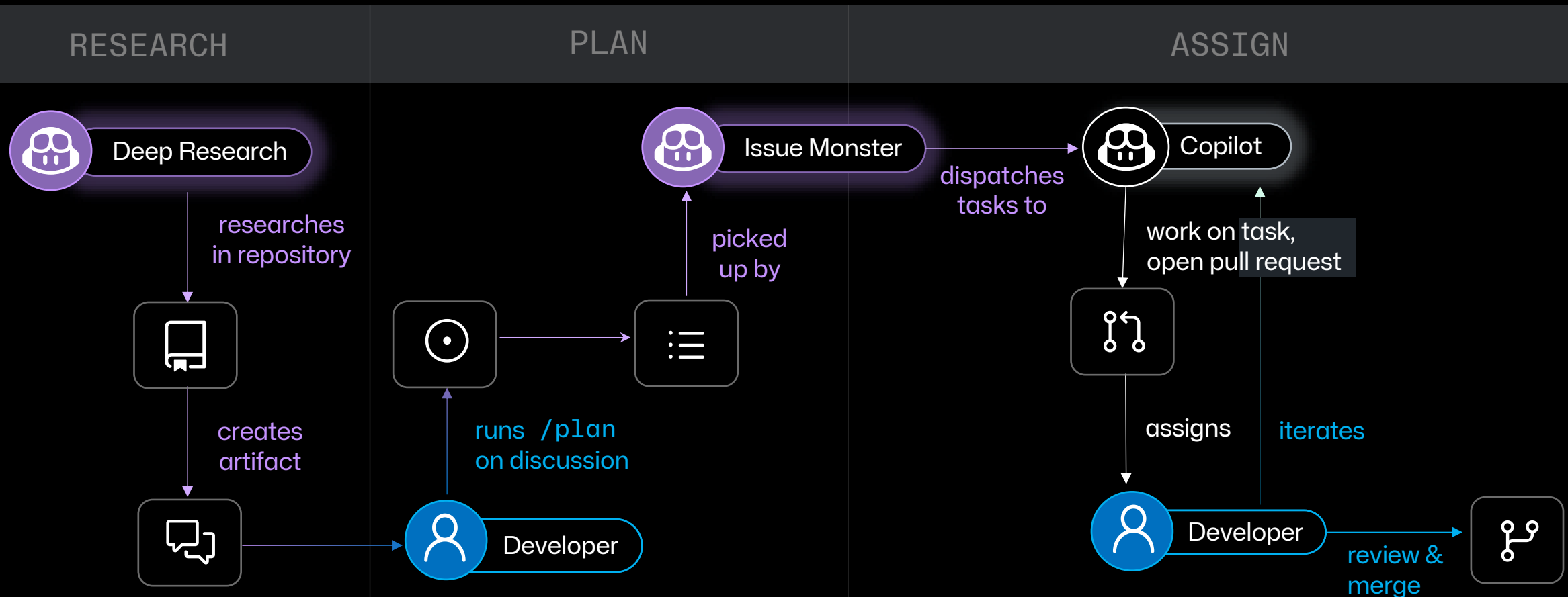
No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

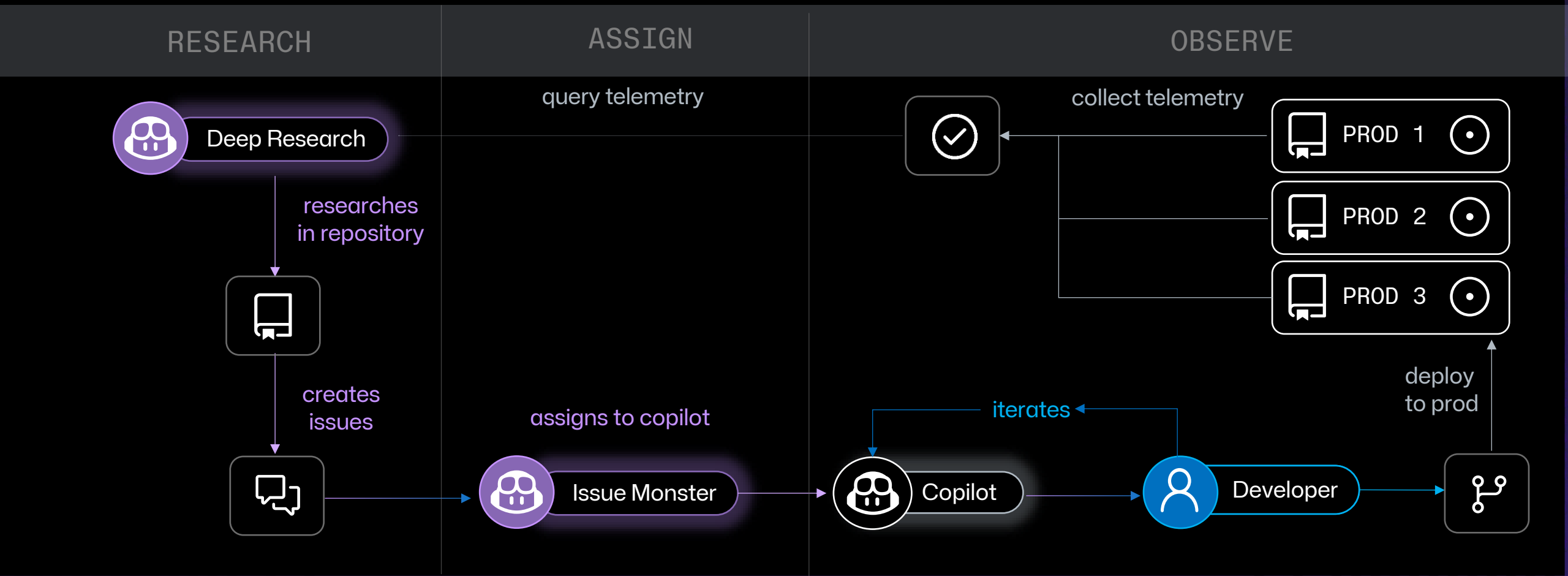
Research – Plan – Assign

Daily Test Improver, Daily Perf Improver, ...



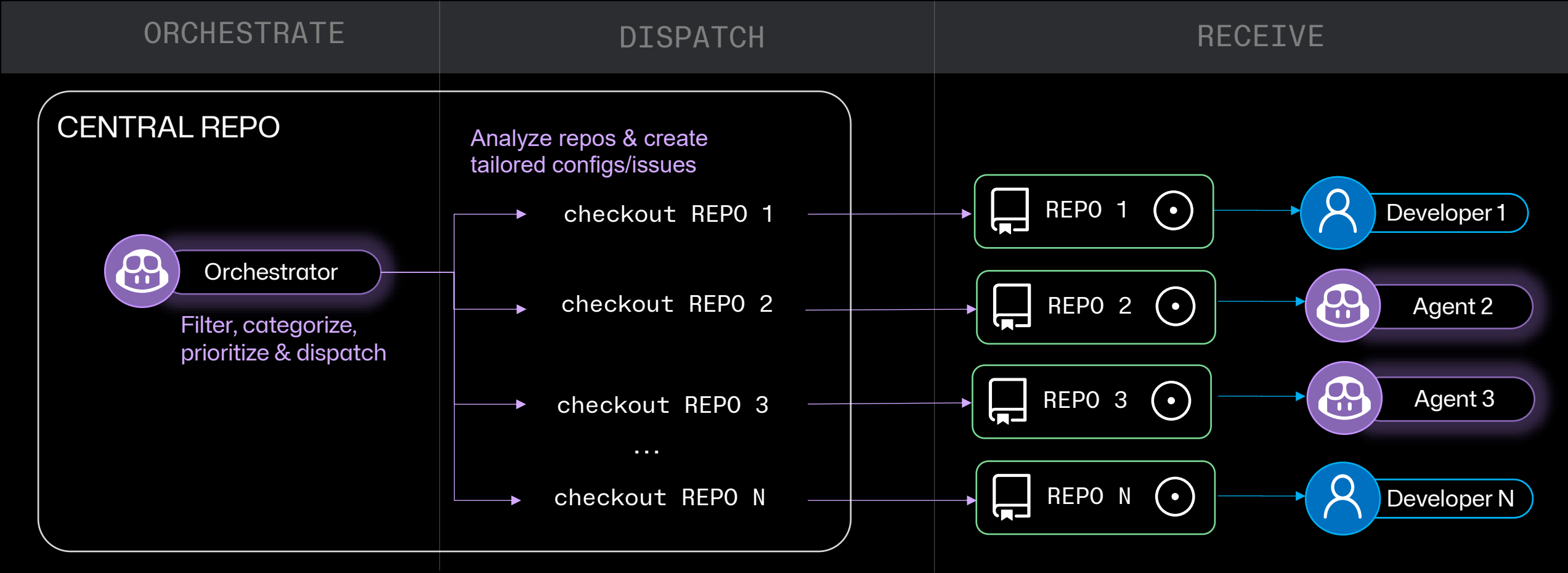
Research – Assign – Observe

Agentic Feedback Loop



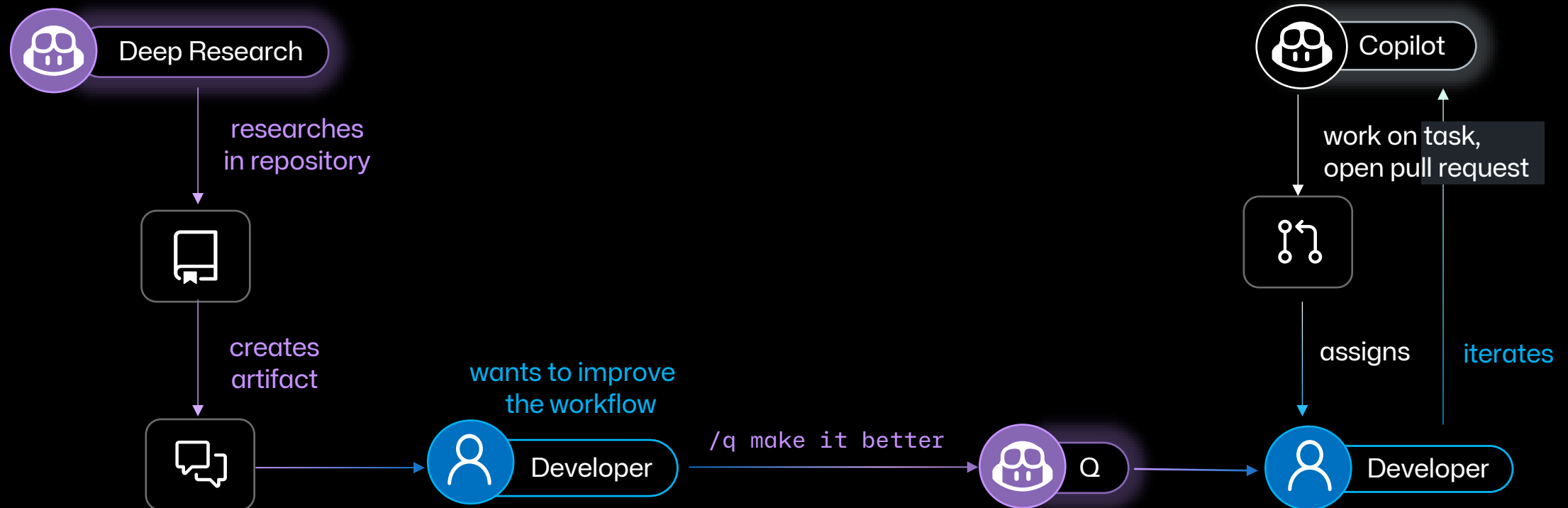
Central Repo Ops

Rolling out large scale operations



Q

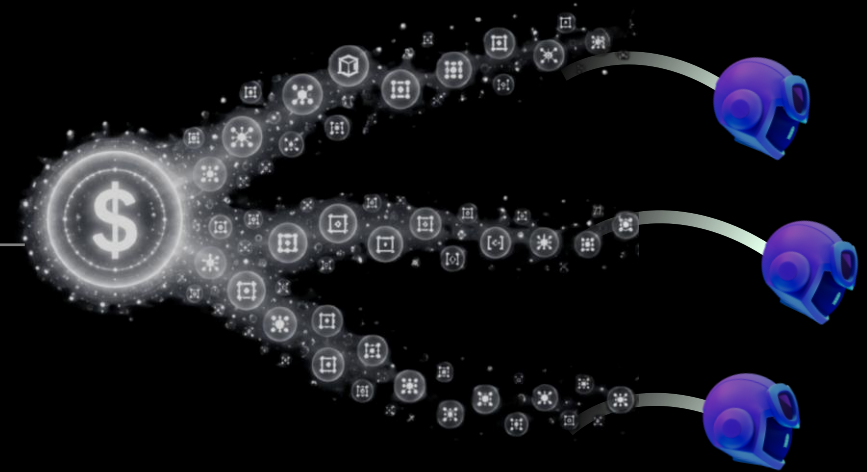
The Agent Improver /q



At scale, agentic AI creates a new problem

The hidden token tax

- Costs accumulate quietly
Agentic AI jobs run automatically.
- Small inefficiencies compound
A few wasted calls per run become expensive.
- Raw token counts can mislead
Model choice, output tokens, workload size, etc.



Make every token traceable

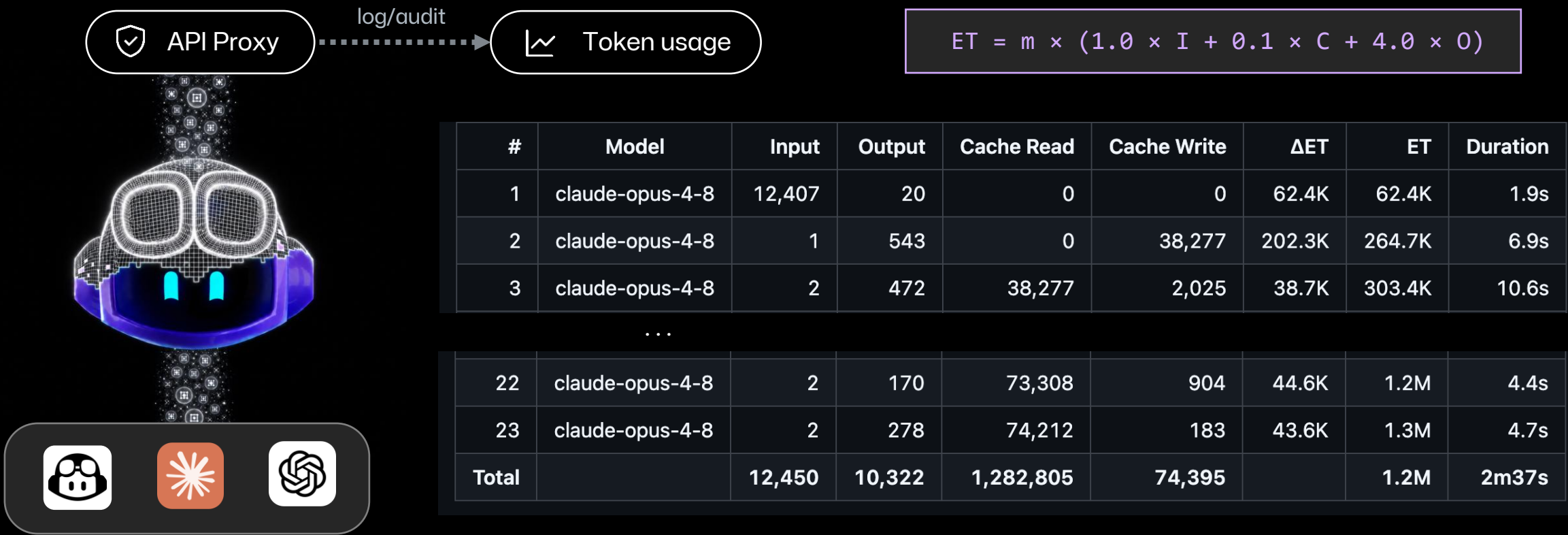


Table: Token usage for a “Smoke Claude” workflow run.

Make every token count

- We use agents to optimize agents
The **Daily Token Usage Auditor** & **Daily Token Optimizer** keep us efficient.



Figure: Daily token usage in the gh-aw repository.

Make every token count



Quick wins

Remove obvious waste

- ☐ Prune unused MCP tools
- ☐ Stop repeated tool calls
- ☐ Fix fallback loops



Structural fixes

Move work out of the agent

- ☐ Pre-fetch diffs, files, metadata
- ☐ Add gates to skip irrelevant runs
- ☐ Use CLI proxies for deterministic reads



Strategic leverage

Optimize the system

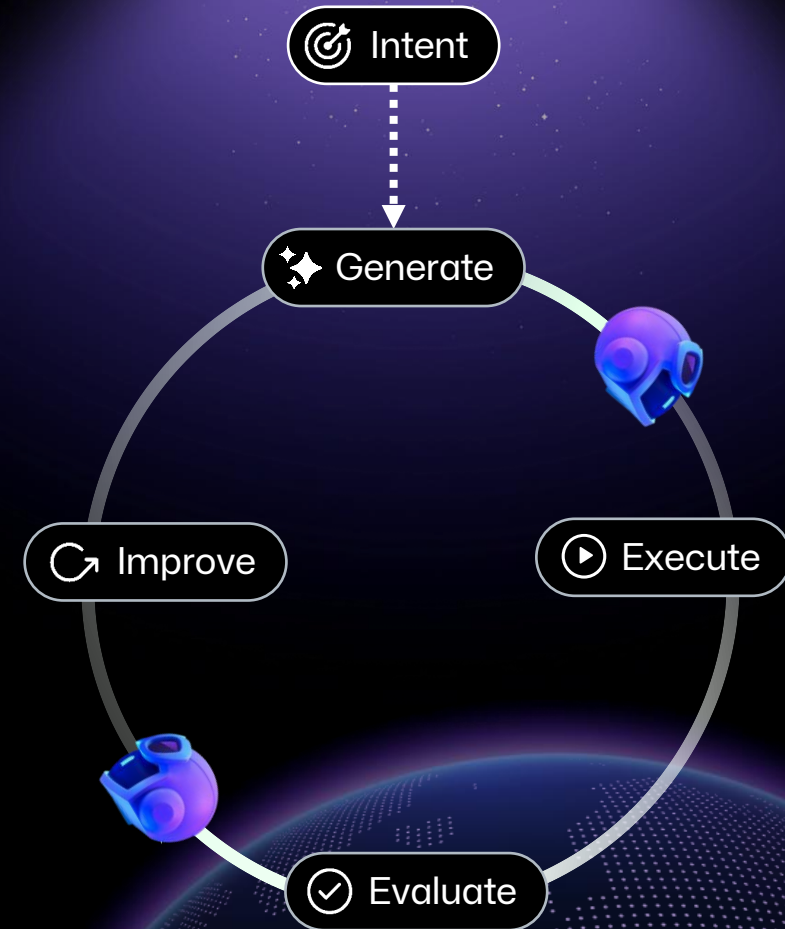
- ☐ Right-size models by task
- ☐ Split monoliths into subagents
- ☐ Run A/B experiments

Make optimization continuous



Add our automatic token audit & optimizers to your repository:

```
gh extension install github/gh-aw  
gh aw add githubnext/agent-ops@v1.0.0
```



Patterns

See our research for more useful design patterns.

gh.io/gh-aw

Welcome to Peli's Agent Factory

Jan 12, 2026



Don Syme



Peli de Halleux



Mara Kiefer

Welcome, welcome, WELCOME to Peli's Agent Factory!

Imagine a software repository where AI agents work alongside your team - not replacing developers, but handling the repetitive, time-consuming tasks that slow down collaboration and forward progress.

Peli's Agent Factory is our exploration of what happens when you take the design philosophy of **“let's create a new automated agentic workflow for that”** as the answer to almost every opportunity that arises! What happens when you **max out on automated agentic workflows** - when you make and use dozens of specialized, automated AI agentic workflows and use them in practice.



